

In-Context Adaptation of Encoder-Decoder Models in Speech Recognition

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Abstract—In-context learning offers an appealing approach to adapt automatic speech recognition (ASR) models to new speakers, accents, and domains by providing speech-text pairs as demonstrations at inference time. Recent work shows that some LLM-based speech models are capable of ASR in-context adaptation, when providing interleaved speech-text demonstrations. In this work, we ask whether in-context adaptation is an inherent ability for all encoder-decoder models. We study two forms of demonstration, collated and interleaved demonstration, across six encoder-decoder models, spanning conventional cross-attention-based and LLM-based architectures. We find that all tested models are able to perform in-context adaptation *out of the box*, achieving up to 30% relative improvement in the oracle experiments and up to 23% using first-pass hypotheses. Through controlled experiments on three English datasets, we show that lexical and speaker information both contribute to successful adaptation. While interleaved demonstration is effective in certain cases, collated demonstration brings consistent adaptation across the board. Our results suggest that in-context adaptation for ASR is not unique to specific architectures, training, or demonstration approaches.

Index Terms—in-context learning, automatic speech recognition, speech LLM, attention encoder-decoder

I. INTRODUCTION

In-context learning (ICL) is the ability to learn new tasks not explicitly seen during training, typically through a few demonstrations (i.e., example-label pairs) provided at inference time. ICL emerged as a special capability in large language models (LLMs) [1], [2], and has since been studied in various speech tasks, such as speech emotion recognition [3]–[5], speech translation [6]–[8], and text-to-speech systems [9], [10], and audio or spoken language understanding [8], [11]–[13]. However, studies of ICL have mostly focused on learning new *tasks* and overlook the potential of using ICL to adapt to different *domains*. If models are capable of learning new tasks at inference time, they should also be able to adapt themselves to unseen domains for the tasks seen during training.

In this work, we study ICL for adaptation, or **in-context adaptation** for short. A model that is able to use the demonstrations provided at inference time to improve performance is said to be capable of in-context adaptation. A model capable of in-context adaptation can, for example, adapt itself to a new accent given the examples of that accent in the demonstrations (accent adaptation) or adapt its output preference given the words in the demonstrations (contextual biasing). We focus on automatic speech recognition (ASR), as it is a

fertile ground for adapting models to new speakers, accents, dialects, and domains. In this work, we restrict our empirical study to English ASR, enabling controlled comparison across models and datasets. Several prior studies have approached ASR adaptation with ICL [14]–[21]. However, prior studies are often limited to a specific model, such as Whisper [14], [15], [17] and Phi-4-multimodal-instruct [19], [21], [22], or suggest that ICL capability is elicited or reinforced through training [17], [18], [23]. Other studies focus solely on finding demonstrations that improve performance [15], [20], [22], [24] but fail to address what exact properties in demonstrations are the most useful for adaptation.

In this work, we study whether modern ASR models can perform in-context adaptation and what properties of demonstrations impact the adaptation the most. Contrary to common belief, demonstrations in ICL for ASR do not necessarily need to be interleaved speech and text. We study two demonstration approaches, **interleaved demonstration** (where a transcript is provided immediately after each of several speech examples) and **collated demonstration** (where all speech examples are provided, followed by all transcripts). We evaluate a diverse set of encoder-decoder models with different architectures and training recipes, including Whisper-large-v2 (Whisper) [25], Canary-1B (Canary) [26], Canary-Qwen-2.5B (Canary-Qwen) [27], Qwen3-ASR-1.7B (Qwen3-ASR) [28], Phi-4-multimodal-instruct (Phi-4-MM) [29], and Qwen2.5-Omni-7B (Qwen2.5-Omni) [30]. We find that all of them are able to perform in-context adaptation *out of the box*, i.e., without any additional training. All models are able to adapt with collated demonstration, whereas adapting with interleaved demonstration is only possible for those that inherit ICL from large language models (LLMs).

We expect the adaptation performance to depend heavily on the demonstrations. To understand what contributes to the adaptation, we study in-context adaptation on L2-Arctic, conducting oracle experiments and controlling the speaker, lexical content, and phonetic content in the demonstrations. We find that sharing both lexical similarity and speaker identity leads to the most performance gains, ranging from 25% to 30% WER reduction across models with collated demonstration. To evaluate in-context adaptation across domains, we extend our analysis to read speech that is less clean (LibriSpeech *test-other*) and spontaneous speech (AMI). Demonstrations sharing similar lexical content still consistently lead to performance

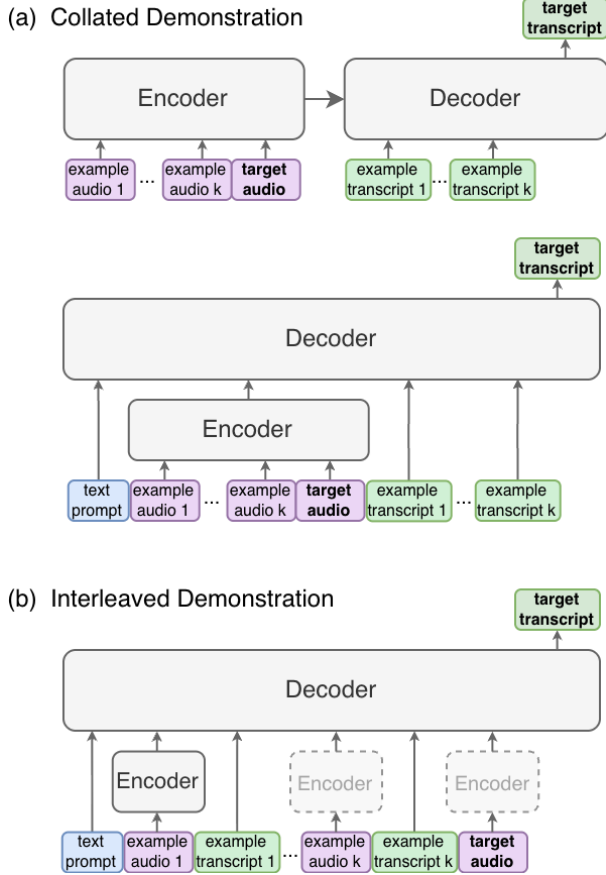


Fig. 1. Two demonstration approaches for in-context adaptation of encoder-decoder models: (a) *Collated demonstration* for AED models (top) and for decoder-only/LLM-based models (bottom) and (b) *Interleaved demonstration* for LLM-based models.

gains, though the limitation of in-context adaptation becomes more obvious.

Finally, to confirm our findings, we relax the oracle experiments by using first-pass hypotheses. The sizable improvements are still present when using first-pass hypotheses, closing up to 60% of the gap against the oracle.

II. COLLATED AND INTERLEAVED DEMONSTRATION

When studying in-context adaptation in ASR, we focus on encoder-decoder models, including models with cross attention (sometimes known as attention-based encoder-decoder models, or AED [25], [31], [32]), and those without (sometimes known as decoder-only models [33], [34]). Recent LLM-based ASR models are of the second kind [18], [28] and still require speech encoders.

As the name suggests, an encoder-decoder model has an encoder (Enc) and decoder (Dec), and ASR is done by the generic equation

$$y = \text{Dec}(\text{Enc}(x)) \quad (1)$$

where y is the output transcript when the model receives the audio x as input.

The input to the decoder is usually more than just the encoded speech $\text{Enc}(x)$, and the additional input is the context that the decoder might adapt to when predicting y . Formally, when using x to predict y , the decoder has access to additional k example recordings $x_1, x_2, \dots, x_k$ and their respective example transcripts $y_1, y_2, \dots, y_k$. We refer to x and y as the target speech and transcript, and the pairs $(x_1, y_1), \dots, (x_k, y_k)$ as **demonstrations**. It is common, especially in text applications, to interleave example-label pairs as demonstrations, but, as we show next, there are potentially better and more natural alternatives.

A. Collated demonstration

Since an encoder-decoder model transcribes speech in sequence, a simple approach to providing demonstrations is, in fact, to present all the example recordings $x_1, \dots, x_k, x$ first followed by all the example transcripts $y_1, \dots, y_k$ and let the model complete the prediction y . Formally, given the k demonstration pairs $(x_1, y_1), \dots, (x_k, y_k)$, decoding with **collated demonstration** for AED models is defined as

$$y = \text{Dec}(\text{Enc}([x_1; \dots; x_k; x]), [y_1; \dots; y_k]), \quad (2)$$

where brackets ($[]$) and semicolons ($;$) are used for concatenating vectors. Fig. 1(a) (top) shows how collated demonstration is done for AED models. The hope is that the additional transcripts $y_1, \dots, y_k$ condition the decoder (to attend to $x_1, \dots, x_k$ using attention) when producing y .

Collated demonstration can be done similarly for decoder-only architectures (such as the LLM-based ones) using the equation

$$y = \text{Dec}\left(\left[\text{Enc}([x_1; \dots; x_k; x]); y_1; \dots; y_k\right]\right). \quad (3)$$

The only difference lies in the final concatenation. Additional text prompts are sometimes necessary for LLM-based models. Fig. 1(a) (bottom) shows how collated demonstration is done for decoder-only architectures.

In this approach, the decoder sees one long recording that is partially transcribed, and is asked to complete the transcription. In other words, this approach is a form of continued decoding or prefix decoding, and is applicable to *any* encoder-decoder models. Collated demonstration has been explored mostly with Whisper [14], [15], [17], [24]. Pan *et al.* [23] is the closest to applying collated demonstration to an LLM-based ASR model, albeit with additional instruction fine-tuning. Some work in text-to-speech apply a similar collated demonstration form for the synthesis to follow the speaker style, prosody, accent, etc., [35]–[37], where they often frame it as in-context learning or conditioning.

B. Interleaved demonstration

Many recent encoder-decoder ASR models are based on LLMs, and part of the reason to build ASR models with LLMs is to inherit their ICL ability. Text-based ICL typically interleaves example-label pairs when constructing demonstrations, and the hope is that LLM-based ASR models can adapt themselves when receiving speech-transcript pairs as context.

Formally, given k demonstrations $(x_1, y_1), \dots, (x_k, y_k)$, decoding with **interleaved demonstration** is defined as

$$y = \text{Dec}([\text{Enc}(x_1); y_1; \dots; \text{Enc}(x_k); y_k; \text{Enc}(x)]). \quad (4)$$

Additional text prompts are almost always needed to instruct the decoder if the model is not trained explicitly on the exact interleaved input of the same task. Fig. 1(b) shows how interleaved demonstration is done for LLM-based ASR models. Models other than those based on LLMs are not intended to be used this way, unless the models are explicitly trained to do so.

Interleaved demonstration has been studied in LLM-based ASR models [18]–[21]. While effective in certain models [18], [29], interleaved demonstration heavily depends on how models are trained and how the text prompts are constructed.

III. EXPERIMENTAL SETUP

We expect the adaptation performance to heavily depend on the relevance between the demonstrations and the speech to be transcribed, and we need settings suitable for controlling phonetic, lexical, and speaker identity. We sample six models and construct controlled demonstration settings on three data sets, with the goal of answering 1) whether the selected models can perform in-context adaptation and 2) what properties of the demonstrations impact adaptation the most.

Results are based on word error rates (WER). We follow the Open ASR Leaderboard [38] and use the Whisper normalizer [25] to normalize hypotheses and references to ensure consistent evaluation. Preliminary experiments show that collated demonstration can be quite sensitive to the text format or the selected examples, so to stabilize the generation, we suppress the EOS token for the first decoding step for all models to ensure the model produces at least one output token for the target audio.

A. Models and Data

We sample six open-sourced models, mainly from the Open ASR Leaderboard, covering attention-based encoder-decoder models (AED) and decoder-only LLM-based models. For LLM-based models, some are omni models that preserve LLM capabilities and can process text, image, and audio, while others behave strictly as an ASR model. Table I summarizes the selected models, their architecture, and whether the model is designed for taking interleaved speech and text. Even though Canary-Qwen and Qwen3-ASR use an LLM, they are not intended to be used for ICL. Phi-4-MM and Qwen2.5-Omni are potentially designed with ICL in mind, since they are exposed to interleaving modality during training and preserve LLM capabilities with speech input.

To control the properties of demonstrations, we evaluate in-context adaptation on L2-Arctic [39], LibriSpeech [40], and AMI [41], [42]. L2-Arctic is a data set consisting of read speech from non-native English speakers. L2-Arctic is ideal for our purpose because the sentences are phonetically balanced and the same set of sentences are read by multiple speakers. L2-Arctic has been recently used to study in-context

TABLE I
A SUMMARY OF THE SELECTED ASR MODELS. MODELS THAT ARE EXPOSED TO INTERLEAVED INPUT MODALITY DURING TRAINING ARE MARKED AS INTENDED FOR ICL.

Model	Architecture	ICL
Whisper [25]	AED	not intended
Canary [26]	AED	not intended
Canary-Qwen [27]	LLM-based (ASR-only)	not intended
Qwen3-ASR [28]	LLM-based (ASR-only)	not intended
Phi-4-MM [29]	LLM-based (Omni)	intended
Qwen2.5-Omni [30]	LLM-based (Omni)	intended

adaptation to speaker, accent, and lexical content [19], [20]. LibriSpeech is commonly used in ASR, but more importantly, it is useful for studying contextual biasing [43]–[45] and in our case, for controlling demonstrations of lexical similarities. AMI consists of spontaneous speech in meetings and is used to test the generalization beyond read speech.

B. Demonstration settings

To study what properties of demonstrations impact the adaptation the most, we consider eight demonstration settings, exploring phonetic, lexical, and speaker similarity. We consider two extreme settings, one with demonstrations randomly sampled from a data set (labeled as **Random**) and one with the demonstration containing the same ground truth transcript as the target (labeled as **Gold text**). We consider two other settings based on phonetic and lexical similarity (labeled as **Phonetic** and **Lexical** respectively). Given an utterance to be decoded, other utterances in the data set are first ranked based on either phonetic similarity or lexical similarity to the current utterance. The first few that are ranked top are used as demonstrations. More formally, given the target speech x and its transcript y , the k demonstrations are collected from a data set S using

$$(x_1, y_1), \dots, (x_k, y_k) = \underset{(x', y') \in S}{\text{top-}k \text{ sim}((x, y), (x', y'))}, \quad (5)$$

where $\text{sim}(\cdot, \cdot)$ is a similarity function. Note that y does not need to be the ground truth transcript of the target speech x and can be the first-pass transcription of an ASR system, whereas the utterances in the data set S always have the ground truth transcript. For simplicity, phonetic similarity is computed based on term frequency-inverse document frequency (TF-IDF) of phone trigrams, and lexical similarity is based on TF-IDF of words excluding stopwords. Designing retrieval methods to select useful demonstrations is an active research area [15], [20], [24], though our choice works sufficiently well as we will see in the experiments. The demonstrations presented to a model are ordered by similarity, with the most similar ordered last, i.e., closest to the utterance to be decoded. All settings are then doubled by controlling whether the demonstrations come from the same speaker or not. Except in the **Gold text** setting, we ensure that the same text as the target is never included in the demonstrations. For the **Gold text** setting with the same speaker, the demonstration is the target speech and its ground truth transcript.

TABLE II

WERs OF IN-CONTEXT ADAPTATION IN CORNER CASES ON L2-ARCTIC. THE SYMBOL $\emptyset$ DENOTES THE BASELINE WITHOUT DEMONSTRATIONS. THE **RANDOM** SETTING INCLUDES A SINGLE RANDOM DEMONSTRATION FROM A *different* SPEAKER, AND THE **GOLD TEXT** SETTING INCLUDES THE TARGET SPEECH AND ITS GROUND TRUTH TRANSCRIPT AS THE DEMONSTRATION.

	$\emptyset$	Random	Gold text
Whisper	7.8	7.3	0.6
Canary	6.8	6.7	2.2
Canary-Qwen	6.7	6.6	0.2
Qwen3-ASR	6.0	5.9	0.1
Phi-4-MM	6.7	6.7	0.3
Qwen2.5-Omni	5.5	5.6	0.1

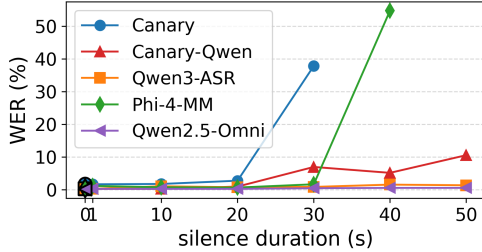


Fig. 2. WERs when separating the demonstration and the target speech with silence in the **Gold text** setting on L2-Arctic. The results show the potential limit of a model’s ability to handle long-form audio.

Even though the **Phonetic** and **Lexical** settings require access to the ground truth transcript of the target utterance, we will relax this assumption and experiment in the same setting with first-pass transcripts to select the demonstrations. We emphasize the importance of these oracle experiments as they answer whether a model is capable of in-context adaptation and to what extent the properties of demonstrations impact the adaptation. The oracle results are also largely missing in prior work.

IV. RESULTS

To see whether the selected models can perform in-context adaptation, we first test the corner cases, providing random demonstrations or gold demonstrations. We will then present a comprehensive comparison including the phonetic and lexical settings, and study generalization of in-context adaptation to other data sets.

A. In-context adaptation in corner cases

There are two corner cases tested here: 1) whether models can ignore irrelevant demonstrations and 2) whether models can copy the ground truth when they are given. For the first case, one irrelevant demonstration is selected randomly from other speakers, and for the second, we provide the identical recording and its ground truth as demonstration. These two corner cases serve as sanity checks that the models need to pass if we want to claim that they can perform in-context adaptation. We limit ourselves to collated demonstration here

because, as we have argued, it is trivially applicable to all models.

Table II shows the results of the two corner cases, and we see that all models pass the test. The results with random demonstrations are close to the results without demonstrations. All models achieve near-perfect WERs when providing the ground truth. Together, all models are capable of in-context adaptation in these two corner cases.

In addition, we insert an arbitrarily long silence between the demonstration and the speech to be decoded. A model that is capable of in-context adaptation should be able to ignore the silence. However, it is well known that encoder-decoder models suffer when the input audio is too long, beyond the typical length seen during training [31], [46], [47]. Fig. 2 shows how models cope with the inserted silence, and we indeed see that Canary and Phi-4-MM (and perhaps Canary-Qwen to a certain degree) can only perform in-context adaptation within a certain context length. For the rest of the experiments, we respect this limit and fit as many demonstrations as possible without breaking the models.¹

B. Impact of different demonstration settings

To quantify how different sources of information in speech contribute to adaptation, we compare demonstrations selected based on phonetic and lexical similarities, in addition to the two corner cases. We also extend the number of demonstrations up to four. After controlling the speakers, we have a total of eight demonstration settings.

Fig. 3 shows how much models are able to adapt based on four demonstrations of different types. First, except **Random** with different speakers, all demonstration settings lead to improved WERs. In particular, all models have no trouble getting near-perfect WERs with the **Gold text** demonstrations. While **Gold text** is not realistic, it does show the importance of speakers in the demonstrations. The **Gold text** WERs are slightly higher when the speakers are different from the target speech.

We then look at demonstration settings that do not contain duplicate text to the target transcript. Selecting random utterances from the same speaker achieves a small but consistent gain. Demonstrations selected based on phonetic and lexical similarity (especially from the same speaker) lead to the most improvement, with lexical similarity slightly better. This trend holds across all models.

Notably, the smaller AED models, Whisper and Canary-1B, achieve similar relative WER reduction to larger models equipped with pretrained LLM decoders. This suggests that the adaptation capability does not inherently depend on the use of a pre-trained LLM.

The amount of adaptation for different numbers of demonstrations is shown in Fig. 4. We only contrast the **Random** and the **Lexical** setting for clarity. Overall, from one to four examples, the more demonstrations we provide, the lower the WER. The first two demonstrations bring the most improvement, and adding more demonstrations diminishes the return.

¹In the case of L2-Arctic, 30 seconds are sufficient to fit 4 demonstrations.

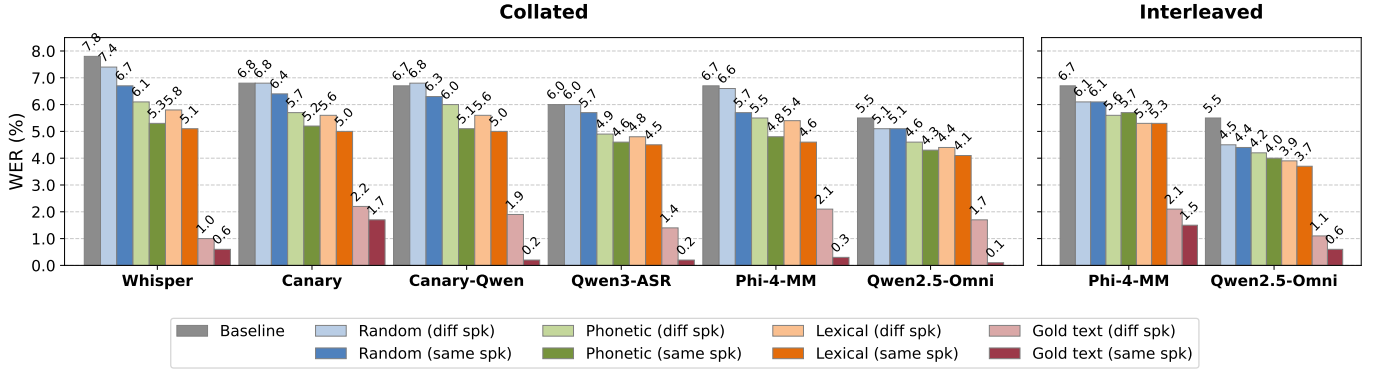


Fig. 3. WERs of in-context adaptation on L2-Arctic for different demonstration settings. We use 4 demonstrations for all settings except the **Gold text**, which only has 1 demonstration.

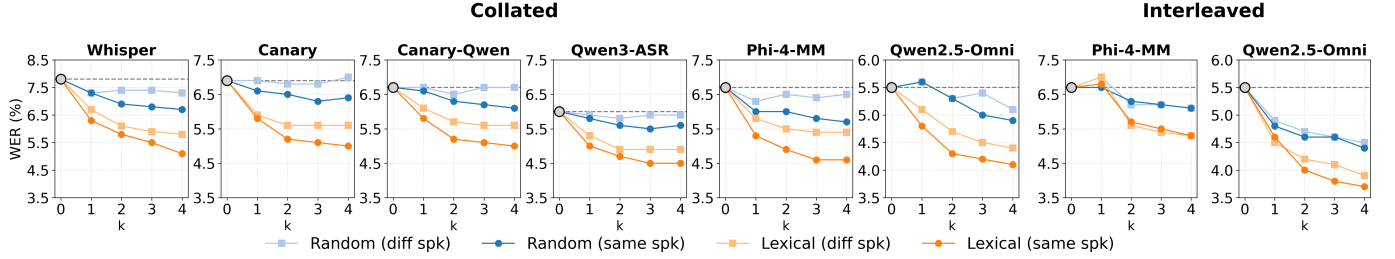


Fig. 4. The *absolute* WER on L2-Arctic under the **Random** and **Lexical** settings as we increase the number of demonstrations from 1 to 4.

C. Comparing collated and interleaved demonstration

Comparing Phi-4-MM and Qwen2.5-Omni in Fig. 3 and Fig. 5, we see that both models are able to adapt based on collated and interleaved demonstration. For interleaved demonstration, we follow the exact same instruction prompt in Roll *et al.* [19] for both Phi-4-MM and Qwen2.5-Omni. Note that prompt engineering is required for interleaved demonstration for both models. Without this instruction prompt, we observe much weaker adaptation, especially if only a single example is provided.

Even though Canary-Qwen and Qwen3-ASR are based on LLMs, they are not intended to take interleaved input and we also do not observe any adaptation effect when providing interleaved demonstrations for these two models. A key advantage of collated demonstration is that it is applicable to all models, requires no modification to the text instruction and task prompt, and works out of the box.

D. In-context adaptation on different domains

We extend our analysis to LibriSpeech (*test-other*) and AMI. We again contrast **Random** and **Lexical** for clarity. The top row of Fig. 5 shows the results on LibriSpeech *test-other*. We see that the overall improvement is smaller on these two data sets compared to L2-Arctic. In particular, providing utterances just from the same speaker is overall less effective. Using lexically similar utterances from the same speaker continues to be the best strategy, achieving at least 20% relative WER reduction with collated demonstration.

Results on AMI are shown in the bottom row of Fig. 5. Since AMI is the more challenging data set consisting of spontaneous speech, the adaptation is again less effective. In fact, Phi-4-MM completely fails under interleaved demonstration, where we found the WER collapses due to severe copying from the demonstration transcripts, though Qwen2.5-Omni adapts just fine. The conclusion is largely the same as in LibriSpeech, lexically similar demonstrations from the same speaker achieve the most improvement.

Interestingly, while LLM-based models show marginal gains under the **Lexical** setting, Whisper shows the largest gains, almost closing the WER gap to the LLM-based ASR models. This again suggests that the effectiveness of in-context adaptation does not necessarily require the use of an LLM.

V. SECOND-PASS IN-CONTEXT ADAPTATION

Our oracle results, though promising, assume access to the ground truth transcript of the target speech. To evaluate the gains in practical scenarios, we relax the assumption and use first-pass transcriptions to find demonstrations of the same speaker and of high lexical similarity. For simplicity, we use the same TF-IDF in the **Lexical** setting in §III-B. We decide to measure similarity using ASR transcripts but still use gold transcriptions in demonstrations. Formally, assuming that we have a data set S with gold transcriptions, we construct the demonstrations with

$$(x_1, y_1), \dots, (x_k, y_k) = \underset{(x', y') \in S}{\text{top-}k \text{ TF-IDF}(T(x), T(x'))}, \quad (6)$$

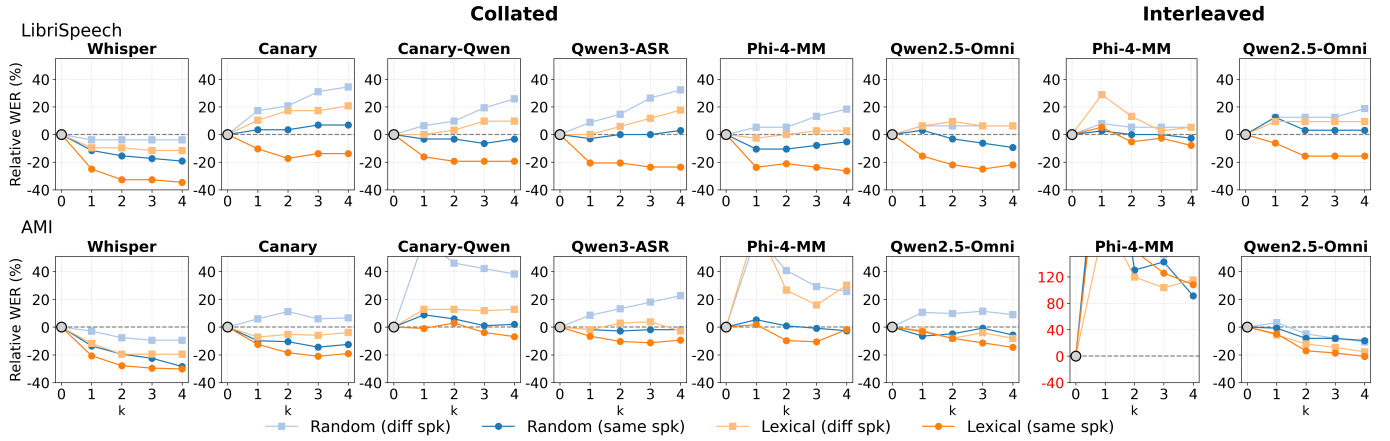


Fig. 5. Relative WER reduction of in-context adaptation against no adaptation on LibriSpeech *test-other* (top) and AMI test split from Open ASR Leaderboard (bottom). We use relative reduction because the WERs span a larger dynamic range across models compared to L2-Arctic.

TABLE III
WERS OF SECOND-PASS IN-CONTEXT ADAPTATION ON L2-ARCTIC IN THE **RANDOM** AND **LEXICAL** SETTINGS USING 4 DEMONSTRATIONS FROM THE *same* SPEAKERS. THE *first pass* COLUMN USES THE FIRST-PASS TRANSCRIPTION FOR SELECTING DEMONSTRATIONS, WHEREAS THE *oracle* COLUMN SELECTS DEMONSTRATIONS WITH THE GROUND TRUTH TRANSCRIPTS.

Model	$\emptyset$	Random	Lexical	
			first pass	oracle
Collated demonstration				
Whisper	7.8	6.7	6.0	5.1
Canary	6.8	6.1	5.5	5.0
Canary-Qwen	6.7	6.2	5.5	5.0
Qwen3-ASR	6.0	5.6	5.0	4.5
Phi-4-MM	6.7	5.7	5.4	4.6
Qwen2.5-Omni	5.5	5.1	4.5	4.1
Interleaved demonstration				
Phi-4-MM	6.7	6.1	5.9	5.3
Qwen2.5-Omni	5.5	4.4	4.2	3.7

where T is an ASR system and $T(x)$ is the first-pass transcript of the target speech x . Note that the ground truth transcript of the example y' is *not* used in computing TF-IDF.

Results with this *second-pass* in-context adaptation on L2-Arctic are shown in Table III. We see a further 3% to 12% relative WER reduction over the **Random** setting, corresponding to closing 25% to 60% of the gaps against the oracle. The second-pass approach is similar to those based on retrieval [15], [20], [24]. While the retrieval approaches are complementary to ours, a simple second pass already brings a significant improvement.

VI. DISCUSSION

Though both collated and interleaved demonstration can achieve in-context adaptation, their underlying mechanism might be different. Collated demonstration operates as one ASR task, while interleaved demonstration requires additional understanding of text instructions and the interleaving format.

This fact makes collated demonstration widely applicable to any encoder-decoder ASR system, and makes interleaved demonstration sensitive to exact wording of the text prompts. Partly because of this, the performance of interleaved demonstration varies a lot more across data sets compared to that of collated demonstration.

Both demonstration approaches, however, are limited by the context length of the model. It is well known that ASR systems suffer from long-form audio or the audio length beyond training [31], [46], [47]. The problem is partially addressed by the use of an LLM, as long-context modeling [29], [48] are now an integrated part of training LLMs. The gains of using more demonstrations do diminish in our experiments, so this limitation might not be as significant.

VII. CONCLUSION

In this work, we study in-context adaptation in ASR with two approaches, collated and interleaved demonstration. We show that, without any training, in-context adaptation achieves significant gains across a diverse set of models and settings.

The oracle experiments not only establish the fact that all models tested are capable of in-context adaptation but also show the limit of what in-context adaptation can achieve. Notably, smaller AED models adapt as effectively as larger LLM-based models, suggesting that in-context adaptation in ASR does not inherently depend on an LLM decoder.

Between the two demonstration approaches, collated demonstration is trivially applicable to any encoder-decoder ASR system, while interleaved demonstration is only applicable to certain models, requiring specific text prompts.

We also show that our approaches are practical, relaxing the oracle results with first-pass hypotheses while maintaining a large proportion of the improvements. Future work includes a deeper understanding of the mechanism behind in-context adaptation, designing experiments to disentangle the contribution of text and audio, developing retrieval approaches for finding better demonstrations, and extending to other settings, such as noise robustness and low-resource ASR.

VIII. GENERATIVE AI USE DISCLOSURE

Generative AI tools (Claude) were only used for language and grammar polishing of the manuscript. All technical content, experimental design, and results were produced by the authors.

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